



AICAS 2022



Tiny-PULP-Dronets: Squeezing Neural Networks for Faster and Lighter Inference on Multi-Tasking Autonomous Nano-Drones

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Why autonomous nano-UAVs?

Huge interest in autonomous (i.e., no external infrastructure) unmanned aerial vehicles (UAVs).
Relevant applications in civil and industrial use cases:

Rescue missions



Surveillance, inspection



Precision agriculture



Entertainment



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Why autonomous **nano-sized** UAVs?

- New use cases: unreachable locations, ubiquitous IoT, etc.
- Reduced costs
- Enhanced safety → e.g., Human Robot Interaction (HRI)

Standard-sized
UAV



Nano-sized
UAV



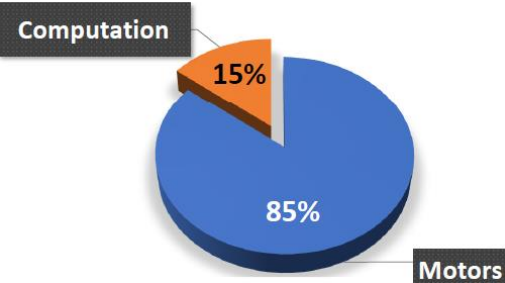
Autonomous nano-UAVs: challenges & goals

Nano-drones challenges:

- Limited payload (~15g)
- Small form factor (~10cm)
- Limited computing power budget (<100mW)

Our goal for autonomous nano-drones:

UAV	Standard-sized	Nano-sized
Size [ø,weight]	~50cm / ~few Kg	~10cm / ~50g
Tot. power	~100 W	~5W
Processing device	High-end CPU	Low-power MCU
Cognitive skills	Fully autonomous & multi-tasking	



R. J. Wood et al., Progress on “Pico” Air Vehicles, 2017.

[1] M. Giurfa and R. Menzel, “Insect visual perception: complex abilities of simple nervous systems” Current Opinion in Neurobiology, vol. 7, no. 4, pp. 505–513, 1997.
[2] P. Foehn et al., “Alphapilot: autonomous drone racing,” Autonomous Robots, vol. 46, no. 1, pp. 307–320, 2022.
[3] R. Bonatti et al., “Towards a robust aerial cinematography platform: Localizing and tracking moving targets in unstructured environments,” IROS, 2019, pp. 229–236.

Autonomous nano-UAVs: challenges & goals

Nano-drones challenges:

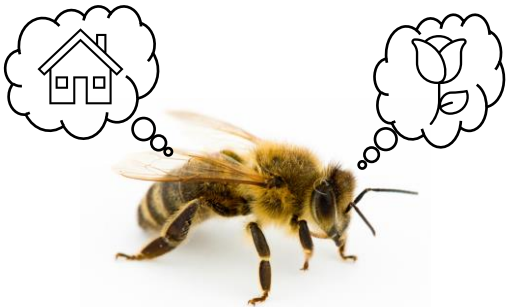
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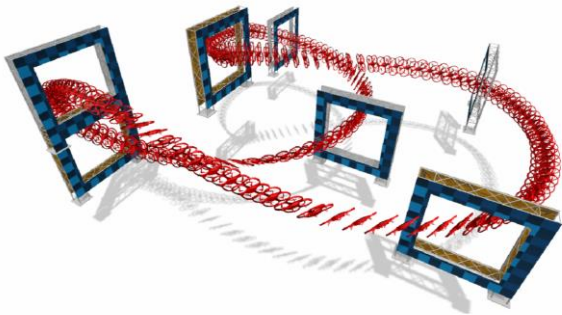
Our goal for autonomous nano-drones:

- Same level of intelligence of standard-sized autonomous UAVs
- Multi-tasking (as biological systems, i.e., honey bees [1])
- Real-time requirements

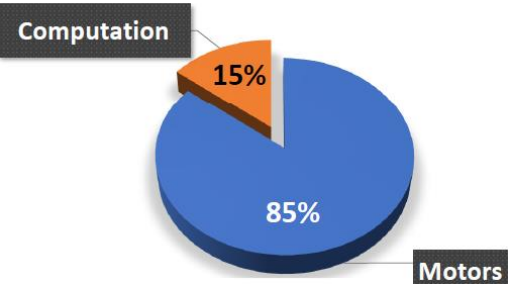
Multi-tasking
biological system [1]



Drone Racing [2]



Aerial photography [3]



R. J. Wood et al., *Progress on "Pico" Air Vehicles*, 2017.

[1] M. Giurfa and R. Menzel, "Insect visual perception: complex abilities of simple nervous systems" *Current Opinion in Neurobiology*, vol. 7, no. 4, pp. 505–513, 1997.

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State-of-the-Art: autonomous nano-drones

SoA limitation: single task intelligence on nano-drones [4,5] (e.g., PULP-Dronet [4] for autonomous navigation)

Memory/computation requirements are still too high for deep learning multi-tasking → we want to enable this possibility

We take the **PULP-Dronet CNN** [4] as use-case to improve.



PULP-Dronet in action.

State-of-the-Art: autonomous nano-drones

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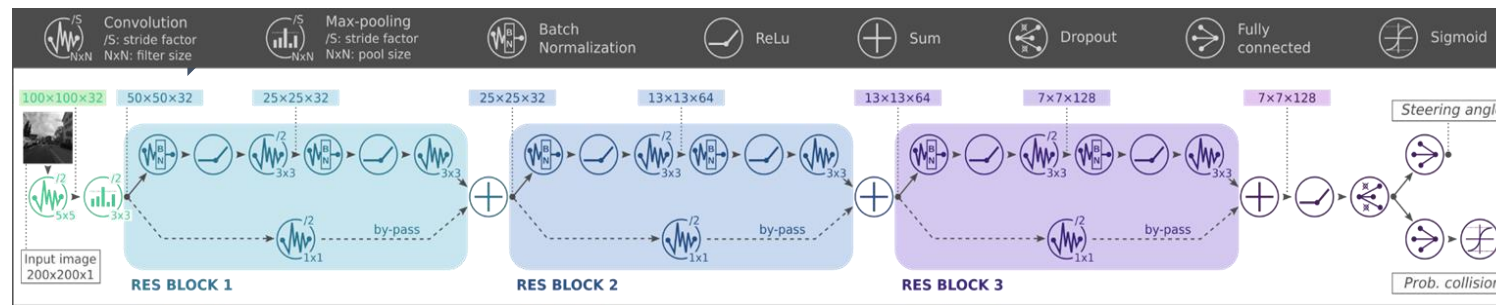
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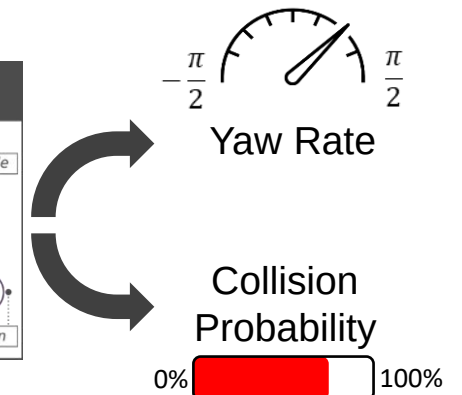
PULP-Dronet in action.

Camera image

PULP-Dronet CNN



Palossi et al. "A 64-mW DNN-based visual navigation engine for autonomous nano-drones." *IEEE Internet of Things Journal*.

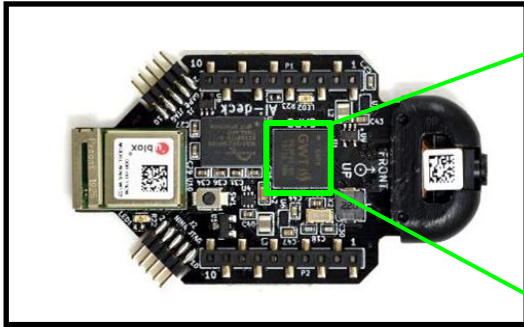


[4] V. Niculescu et al., "Automated Tuning of End-to-end Neural Flight Controllers for Autonomous Nano-drones." *AICAS'21*, pp. 1-4. IEEE, 2021.

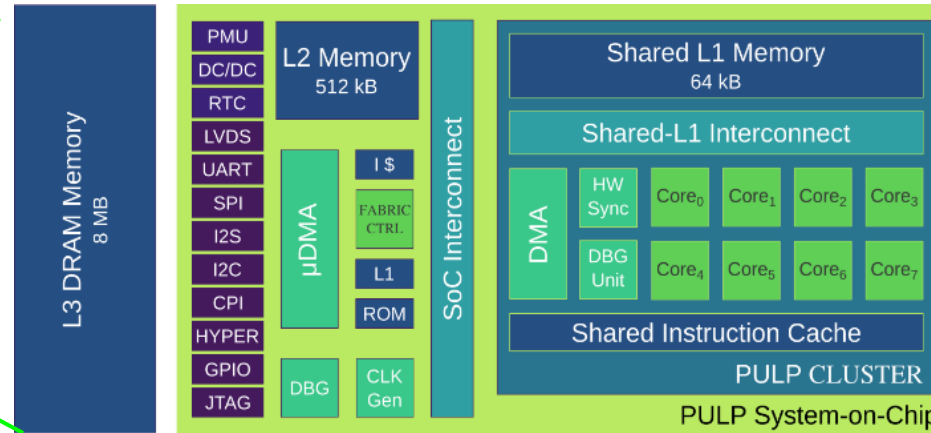
[5] D. Palossi et al., "Fully Onboard AI-Powered Human-Drone Pose Estimation on Ultralow-Power Autonomous Flying Nano-UAVs," in *IEEE Internet of Things Journal*, vol. 9, no. 3, pp. 1913-1929

How to enable multi-tasking on nano-drones

AI-deck



GreenWaves Tech. GAP8 System-on-Chip



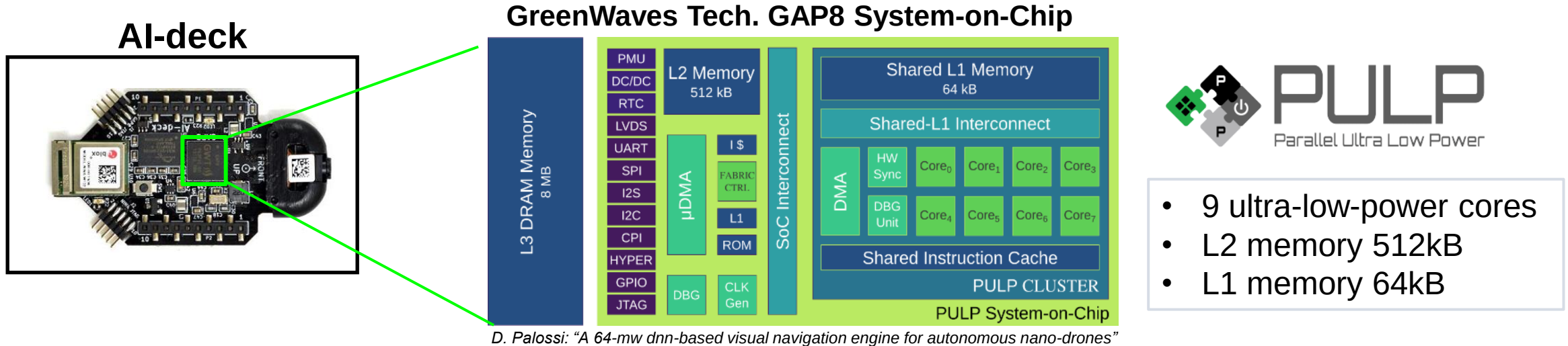
- 9 ultra-low-power cores
- L2 memory 512kB
- L1 memory 64kB

D. Palossi: "A 64-mw dnn-based visual navigation engine for autonomous nano-drones"

Reduction of *i.)* the memory footprint and *ii.)* number Multiply-Accumulate (MAC) operations.

We present a general methodology to study the size and complexity trade-offs on neural networks that suffer from *i) sparsity* and *ii) overfitting*.

How to enable multi-tasking on nano-drones



Reduction of *i.)* the memory footprint and *ii.)* number Multiply-Accumulate (MAC) operations.

We present a general methodology to study the size and complexity trade-offs on neural networks that suffer from *i) sparsity* and *ii) overfitting*.

We apply our methodology to PULP-Dronet to enable **multi-tasking on resource-limited MCUs** (e.g., our nano-drone) by increasing the throughput and reducing the memory footprint.

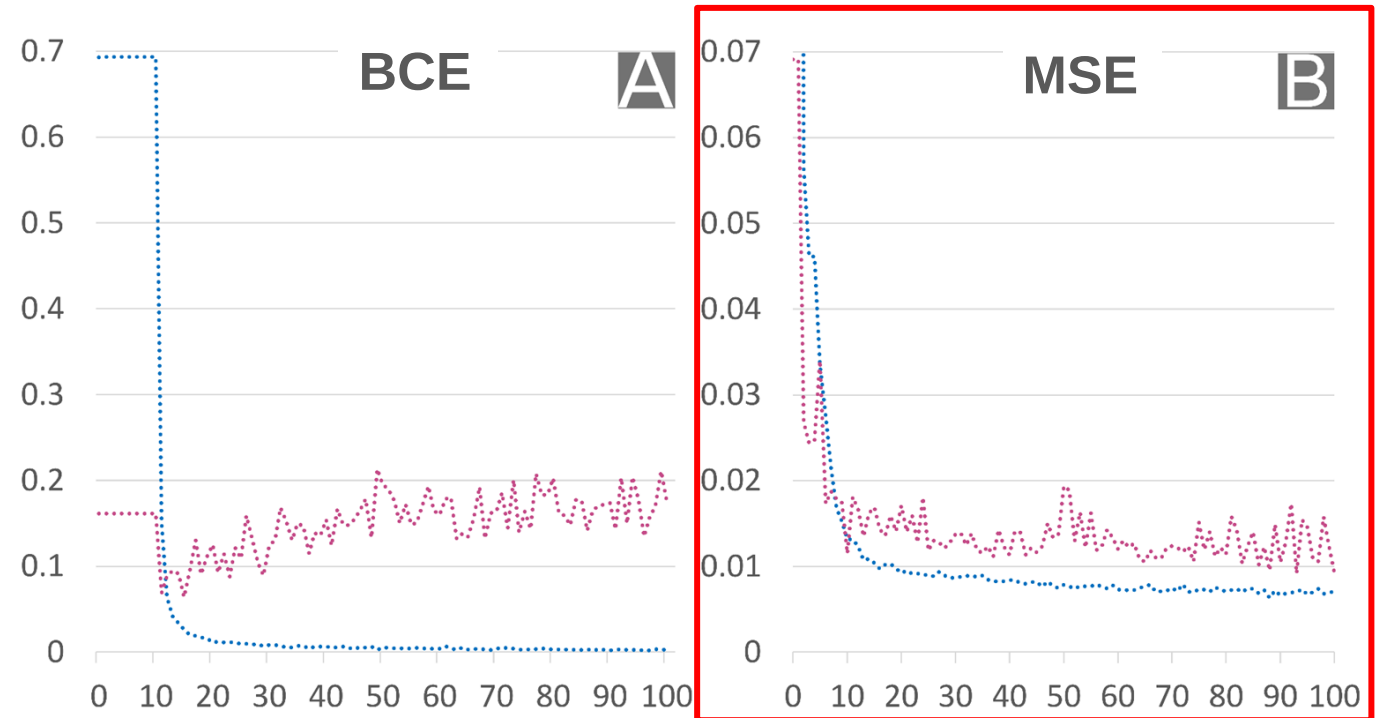
Methodology: overfitting analysis

Dotted lines = Baseline

Solid lines = Tiny-PULP-Dronet

Baseline model validation:

- MSE= noisy but constant



Baseline Model Training Validation
Tiny Model — Training — Validation

BCE = Binary Cross Entropy (collision output)

MSE = Mean Squared Error (steering output)

Methodology: overfitting analysis

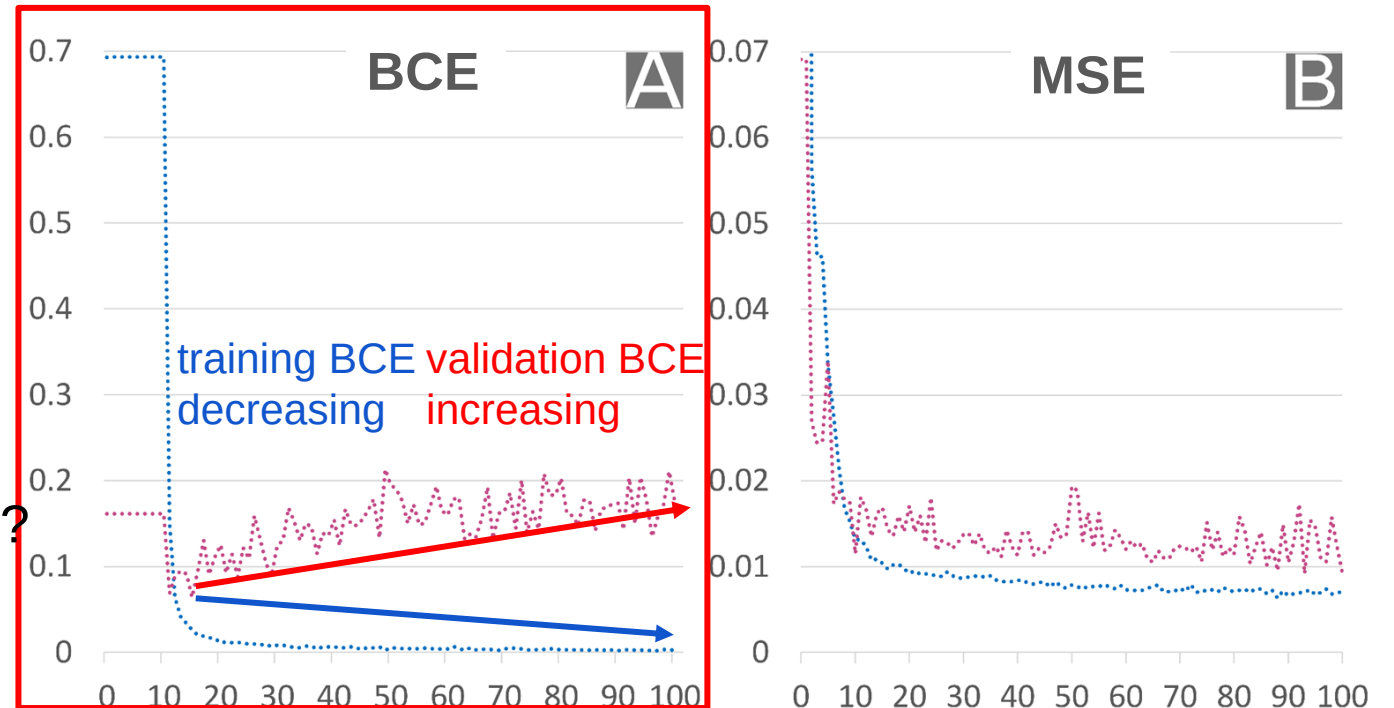
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- BCE = increasing error → **Overfitting**

BCE overfitting: overparametrized model?



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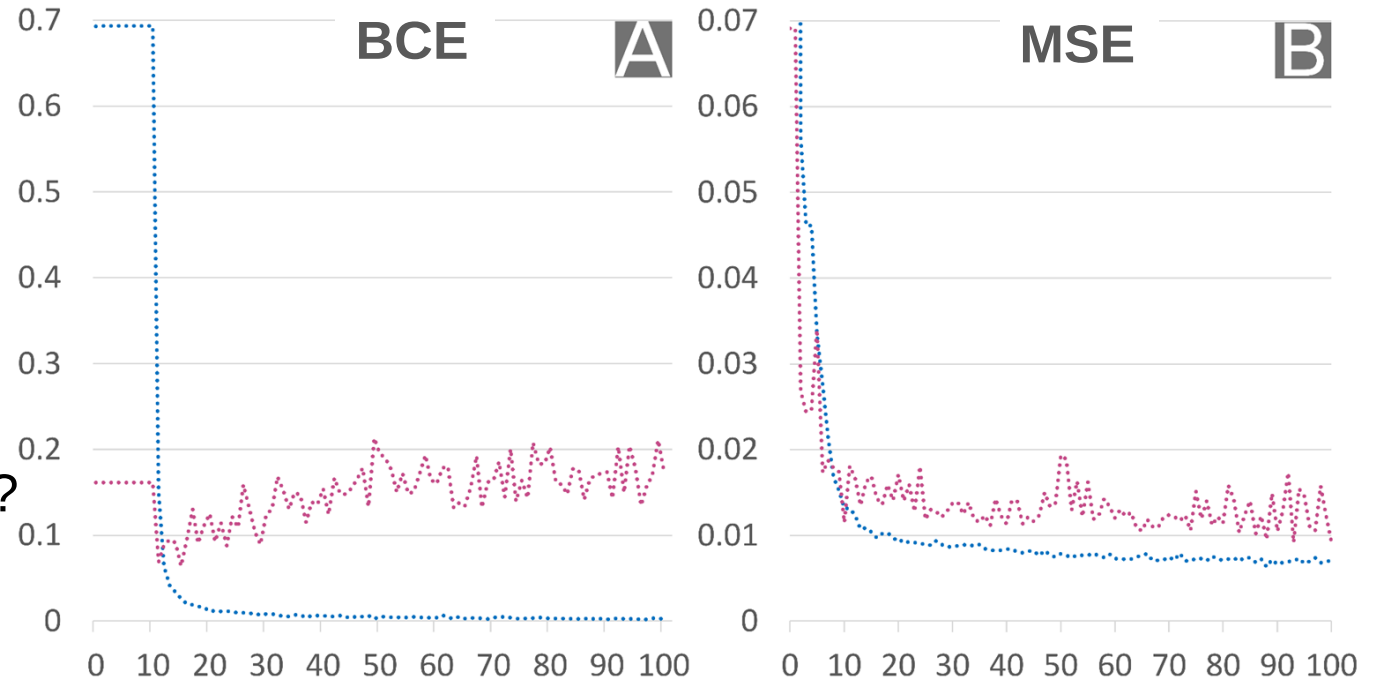
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Model shrinking: thinning the network
applying a scaling factor γ to all channels



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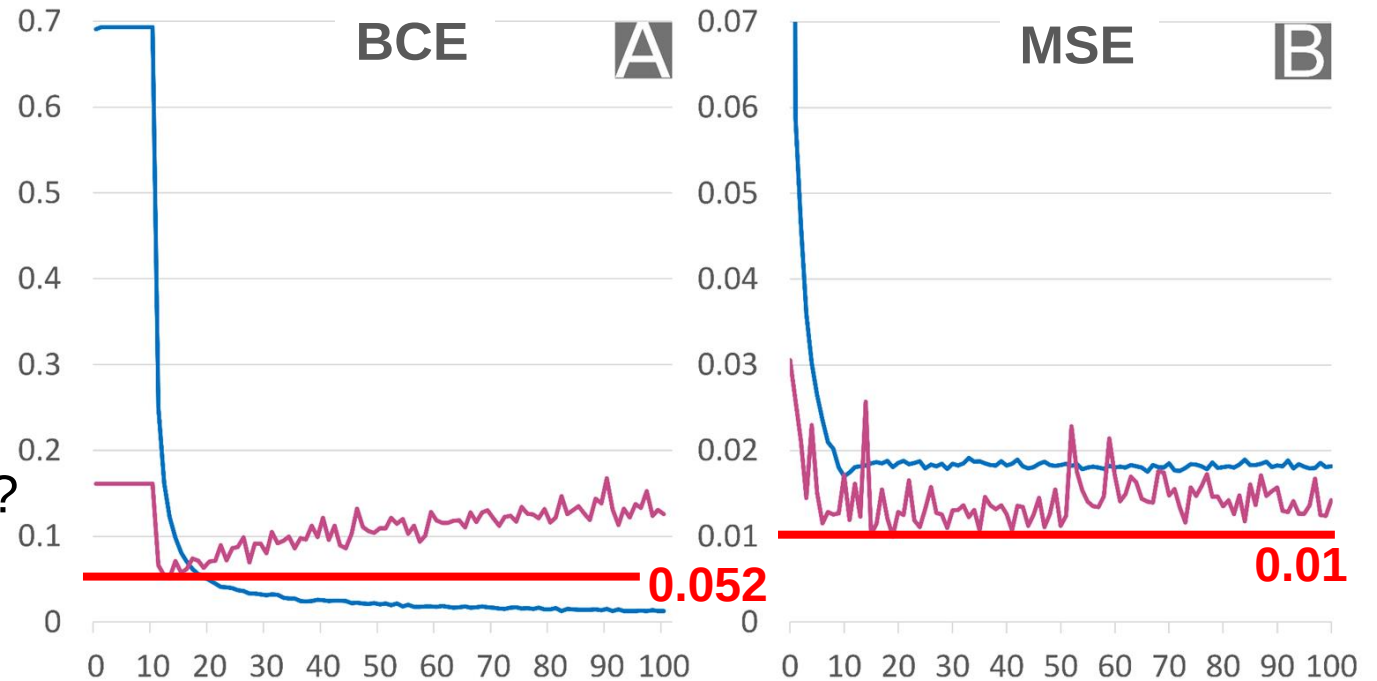
- MSE= noisy but constant
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BCE overfitting: overparametrized model?

Model shrinking: thinning the network applying a scaling factor γ to all channels

Outcome: Tiny model ($\gamma=0.125$) reaches

- same minimum MSE error (0.01)
- reduced BCE error (0.052 vs. 0.064)



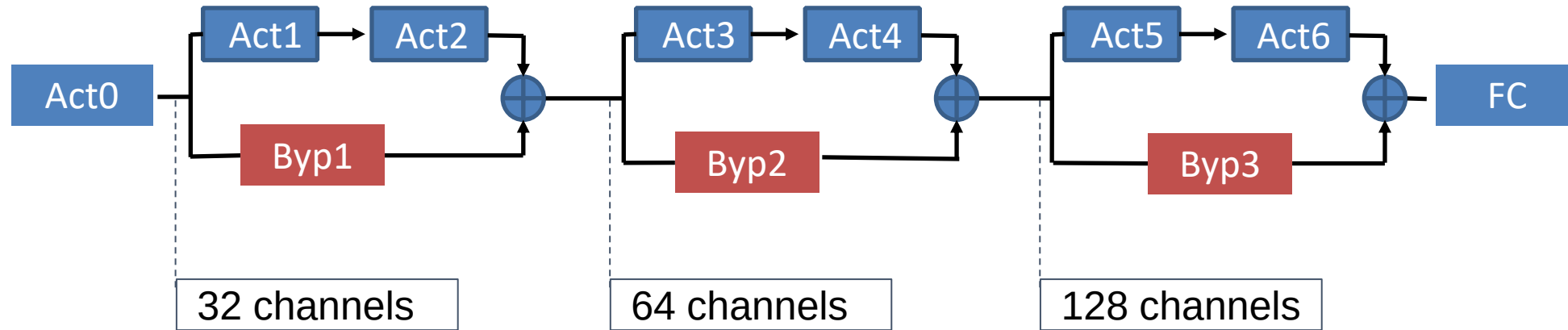
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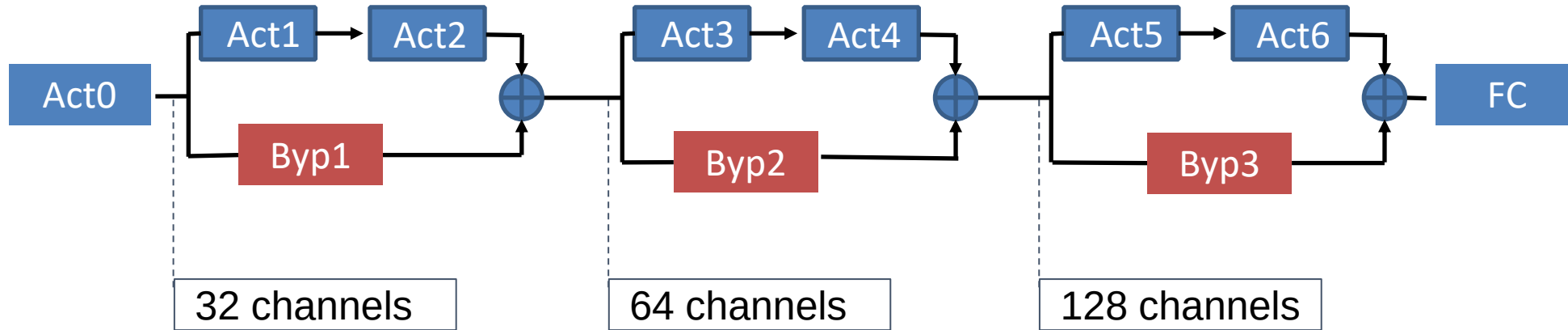
Model shrinking approach

Thinning the network applying a scaling factor γ to all channels



Model shrinking approach

Thinning the network applying a scaling factor γ to all channels



Analysis on the following configurations:

- 32 - 64 - 128 channels (1x) (*Baseline*)

Tiny-PULP-Dronets:

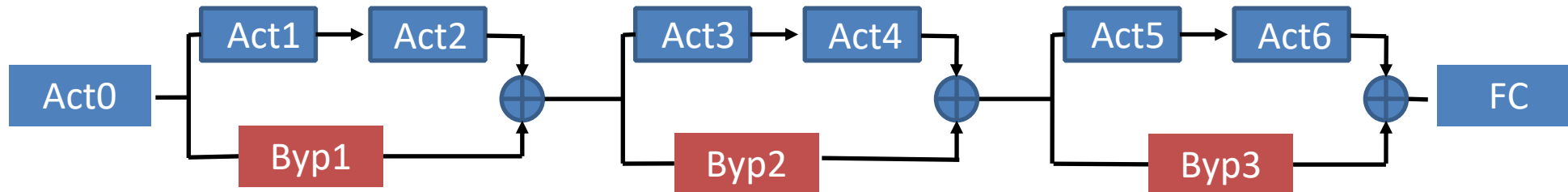
- 16 - 32 - 64 channels (0.5x)
- 8 - 16 - 32 channels (0.25x)
- 4 - 8 - 16 channels (0.125x) (*Smallest*)

Methodology: sparsity analysis

note: structural $\neq$ structured

Structural activation's sparsity accounts for neurons that are never activated (always=0) – input independent.

How we use it: a tool for identifying and pruning neurons not contributing to the learning process [6].



Structural activation's sparsity on PULP-Dronet:

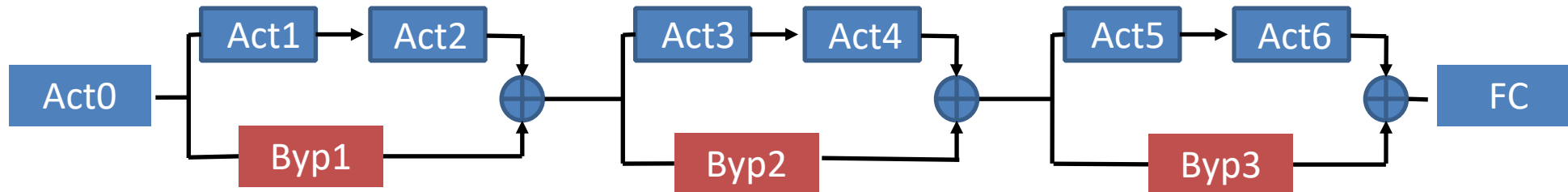
	Conv	ResBlock1			ResBlock2			ResBlock3		
Topology	Act0	Act1	Act2	Byp1	Act3	Act4	Byp2	Act5	Act6	Byp3
Baseline	28%	13%	6%	6%	25%	10%	11%	49%	60%	92%

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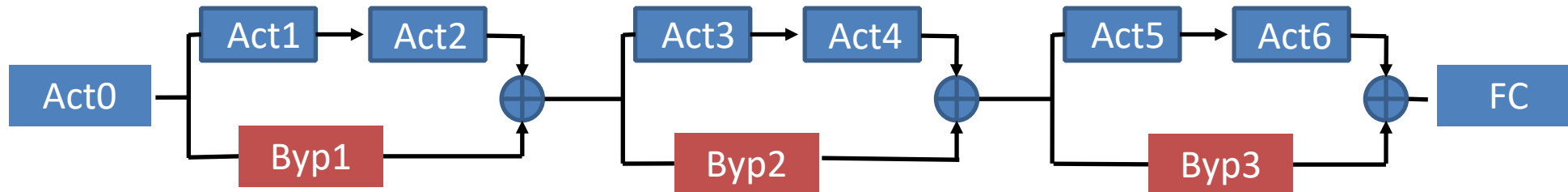
**92% of neurons
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By-pass branches: introduced with ResNets, useful for avoiding vanishing gradients in very **deep networks**.

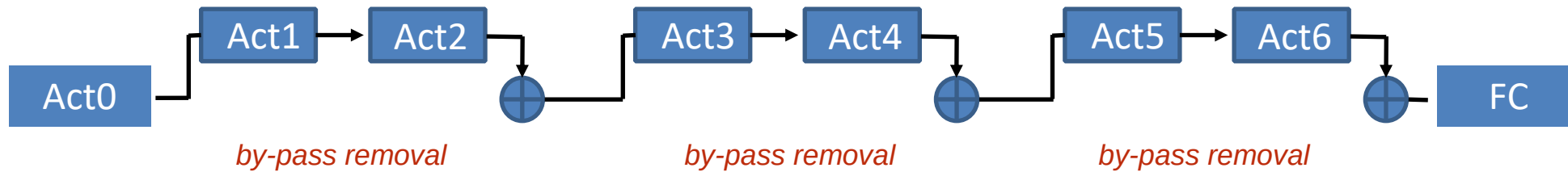
- But PULP-Dronet has only 11 layers → **very shallow**

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Baseline	28%	13%	6%	6%	25%	10%	11%	49%	60%	92%
Tiny	0.15%	0%	0%	-	0%	0.07%	-	0%	0%	-

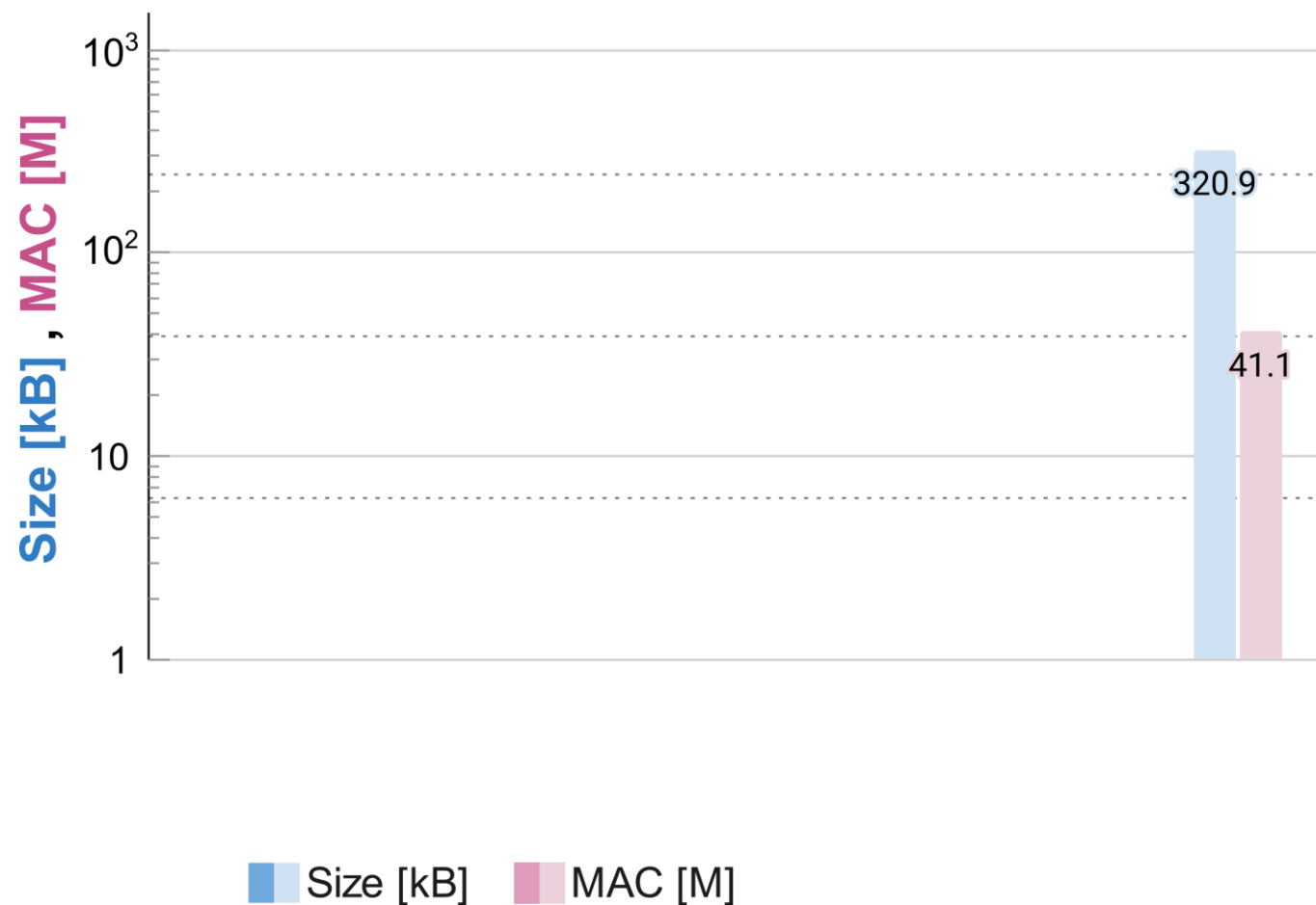
No vanishing gradients: average gradient magnitude = 0.055 (worst case)

Conclusion: the network pruning -- **removing the by-pass branches** -- allows a more efficient usage of the neurons left.

Results: model evaluation

Baseline PULP-Dronet:

- Size: 320kB
- MAC: 41M

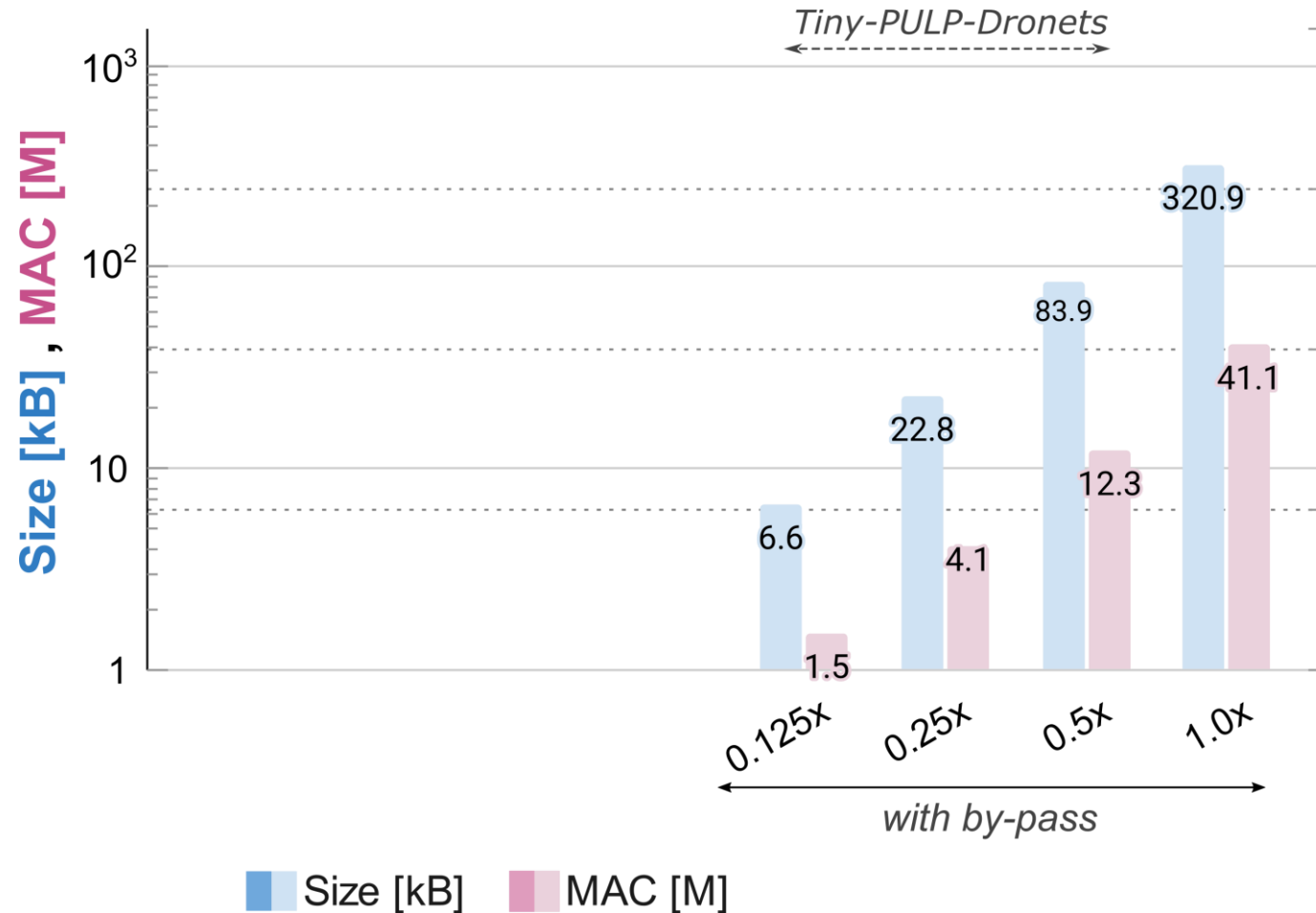


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Tiny-PULP-Dronets: we apply both
1. model shrinking



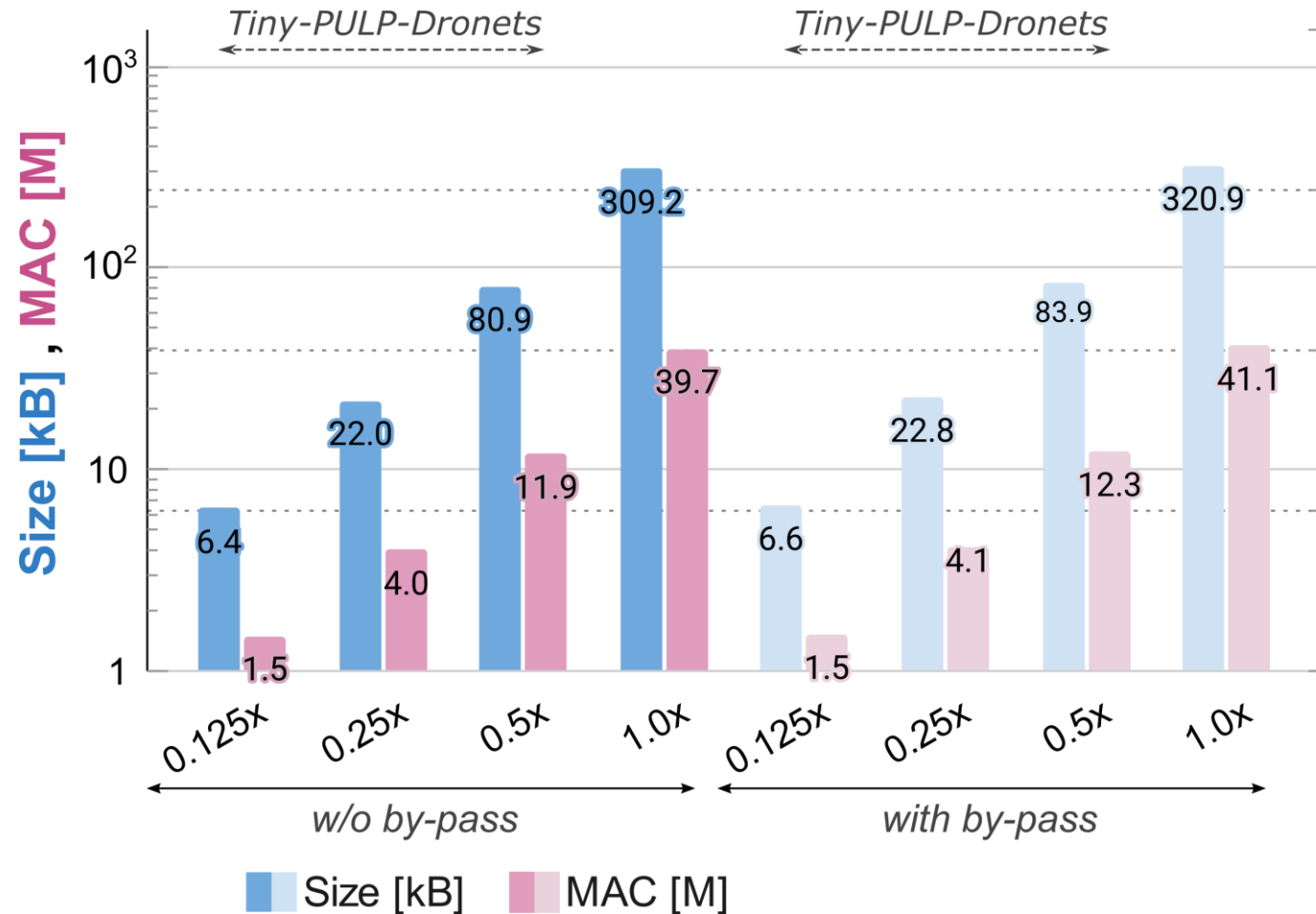
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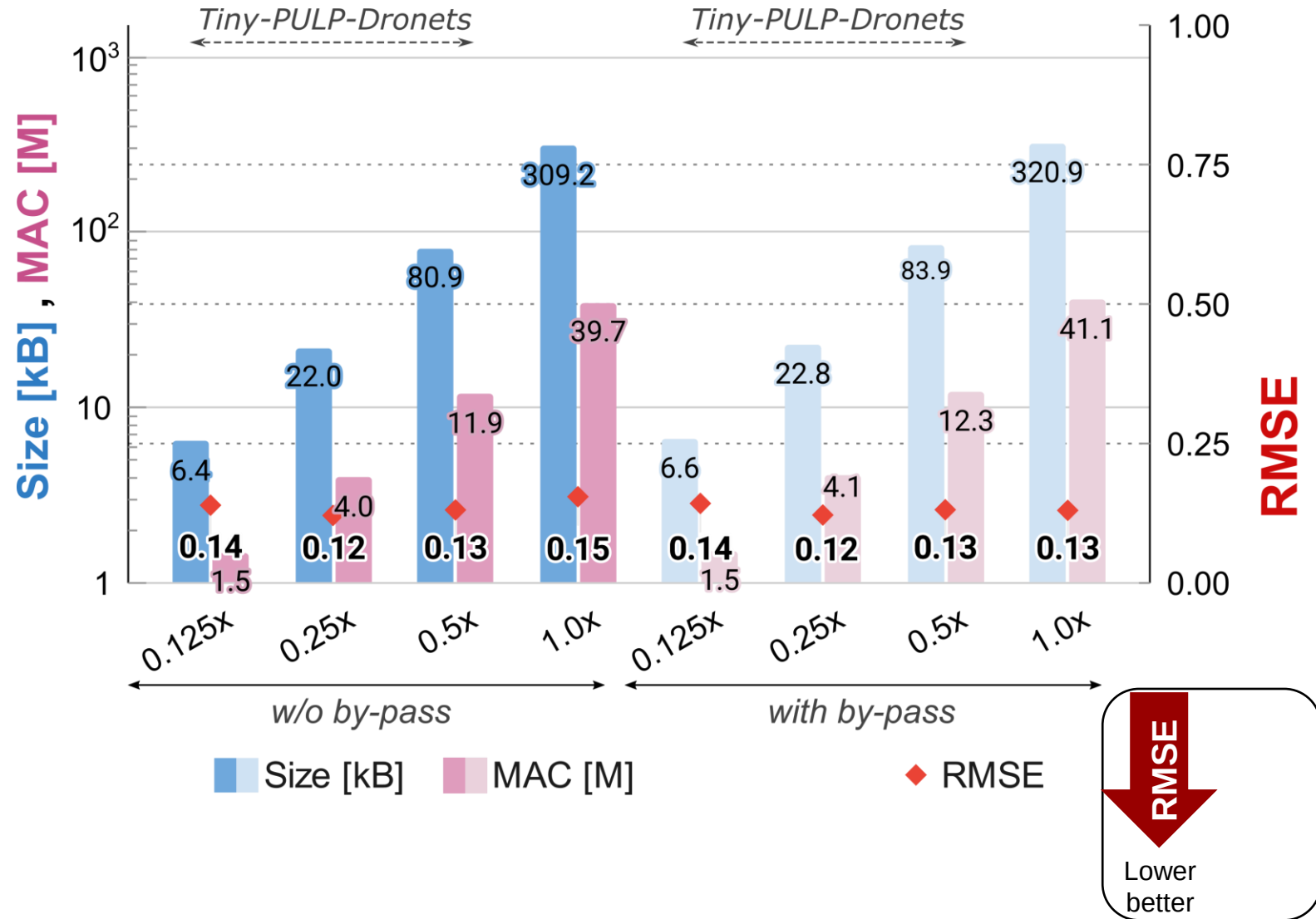
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We compare: **RMSE**



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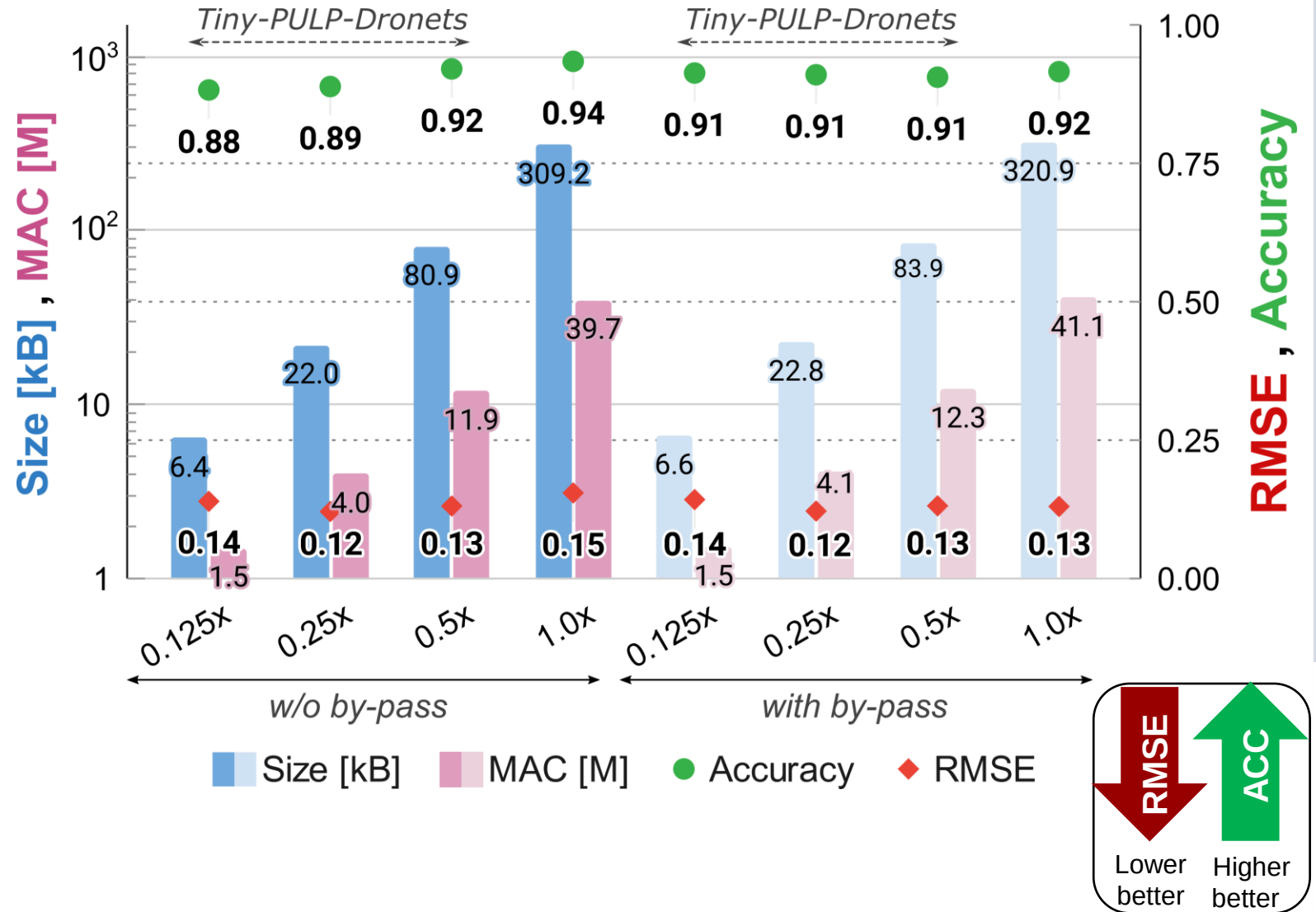
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We compare: RMSE and **Accuracy**



Results: model evaluation

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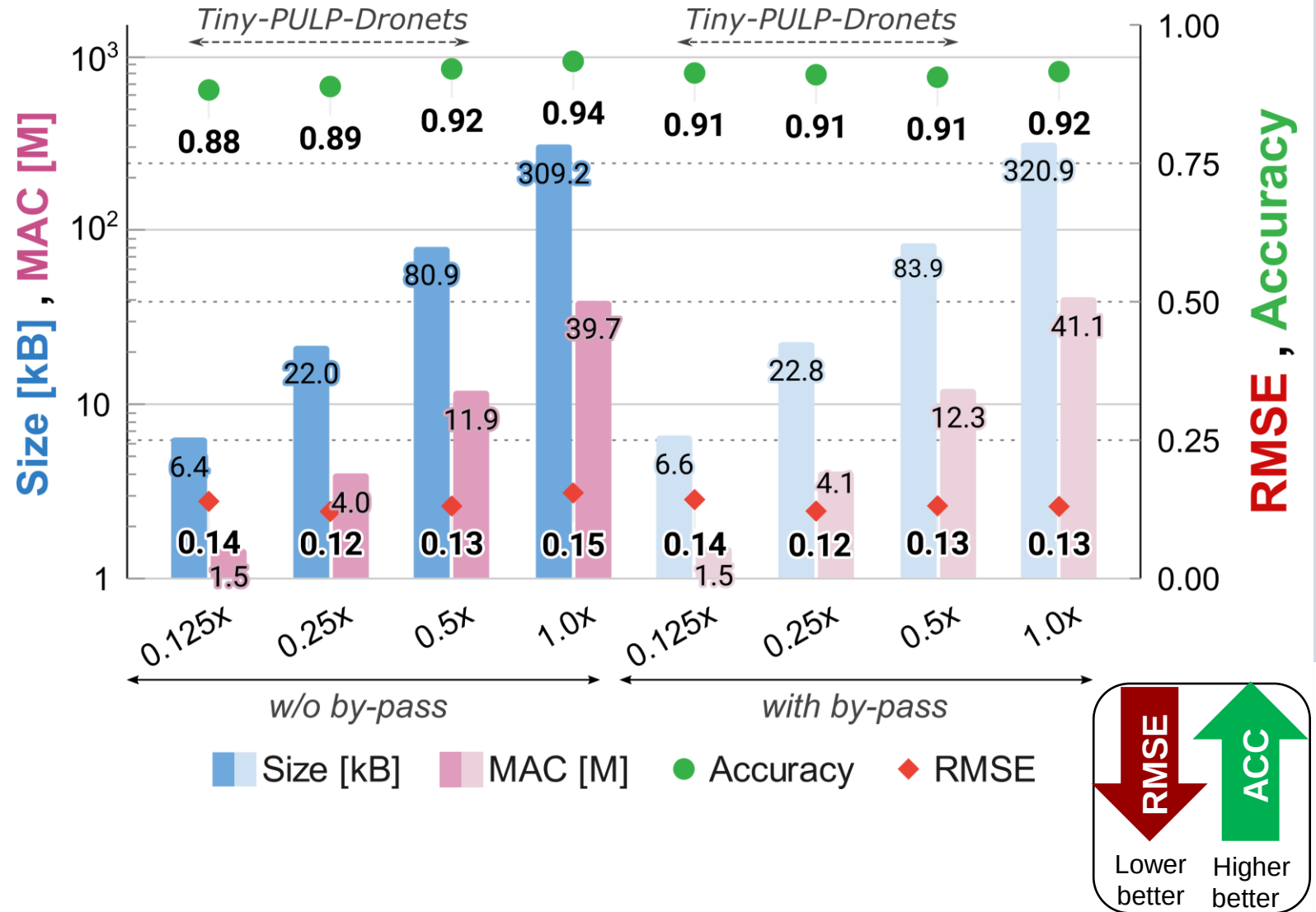
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Tiny-PULP-Dronets: we apply both

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We compare: RMSE and Accuracy.
Not significantly impacted:

- Acc -4%, RMSE +0.01



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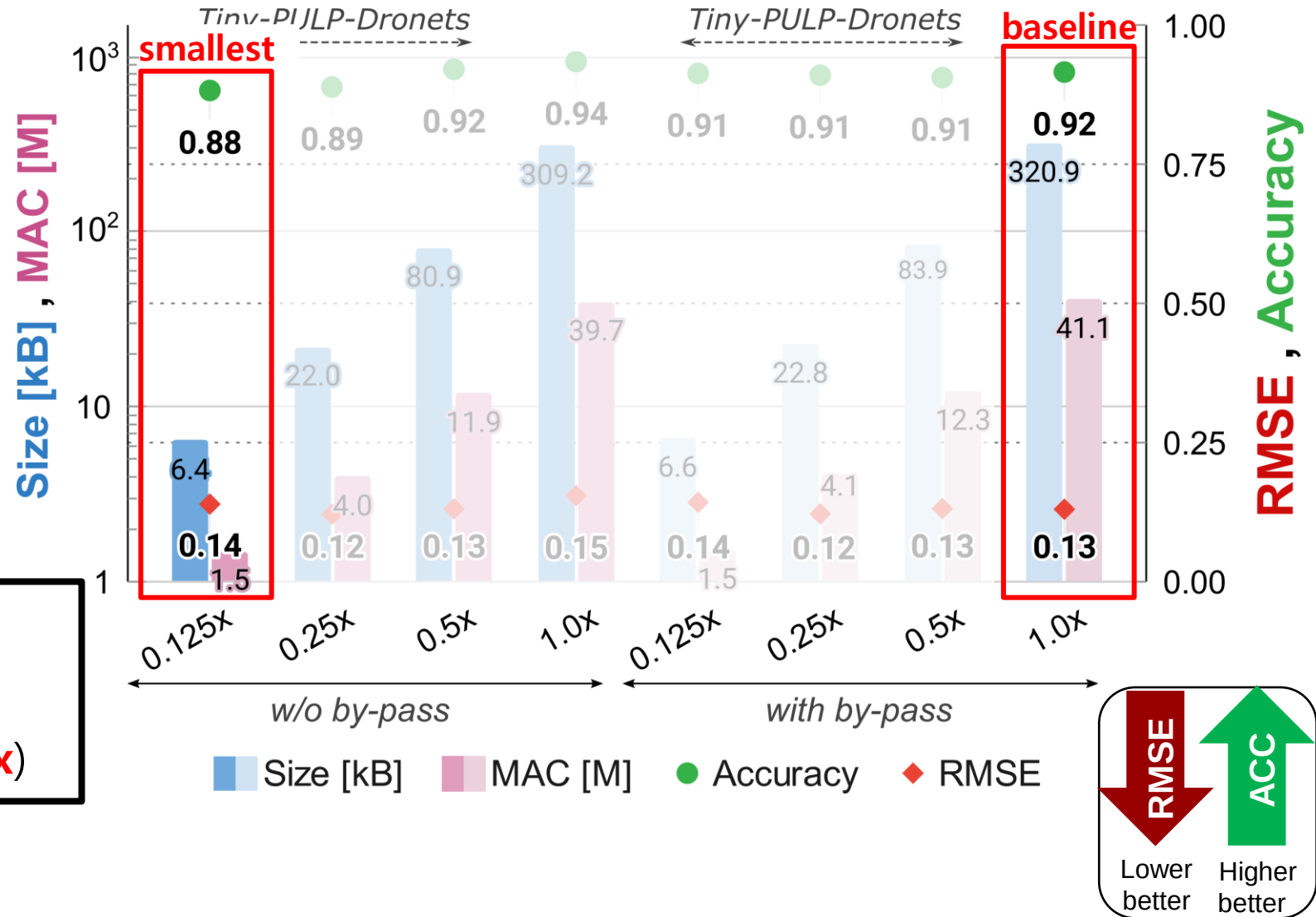
We compare: RMSE and Accuracy.

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Result: smallest vs. baseline

- Size: from 320kB to 6kB (**50x**)
- MAC: from 41M to 0.85M (**27x**)



Results: model evaluation

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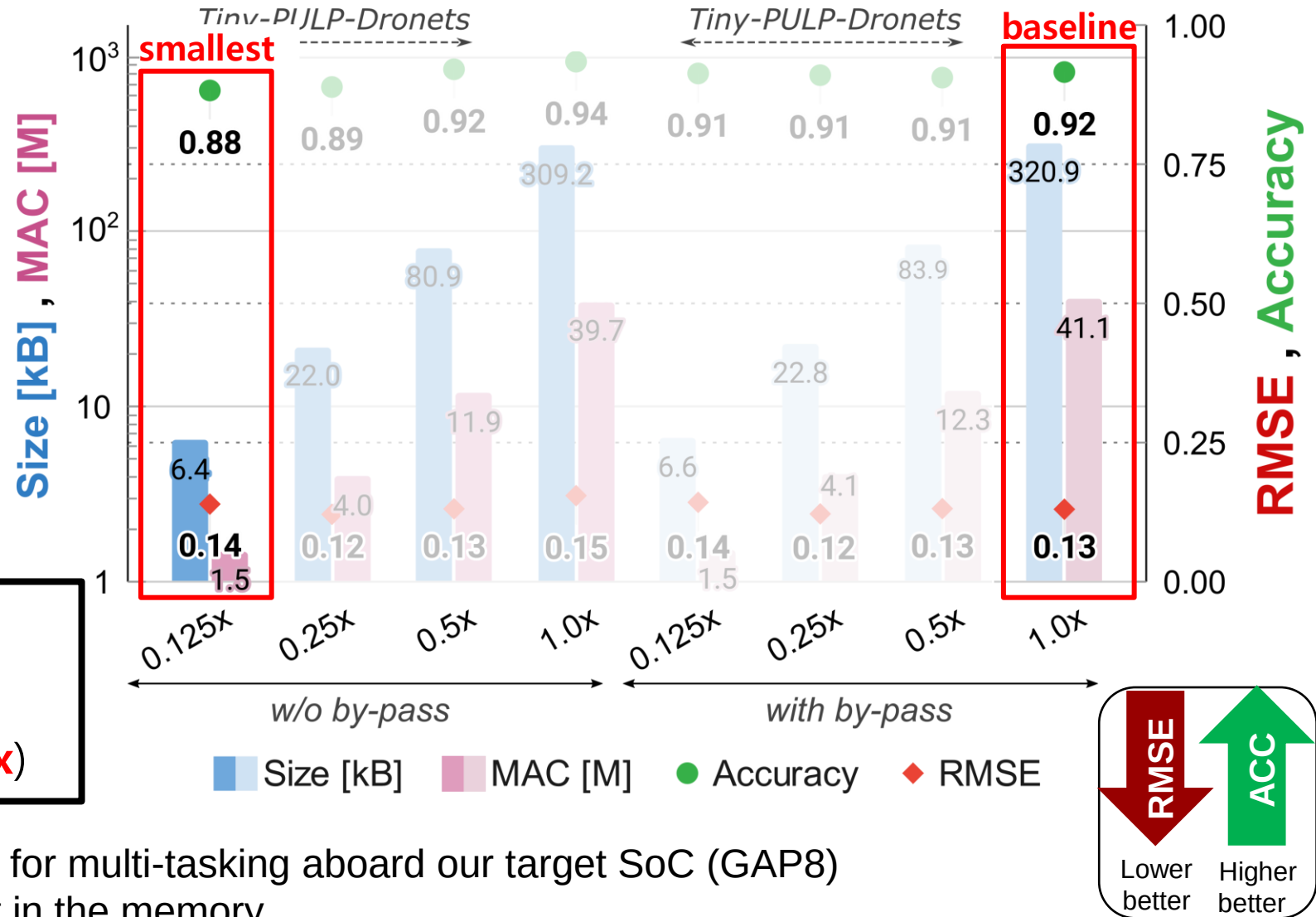
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Result: we can leave plenty of space for multi-tasking aboard our target SoC (GAP8)

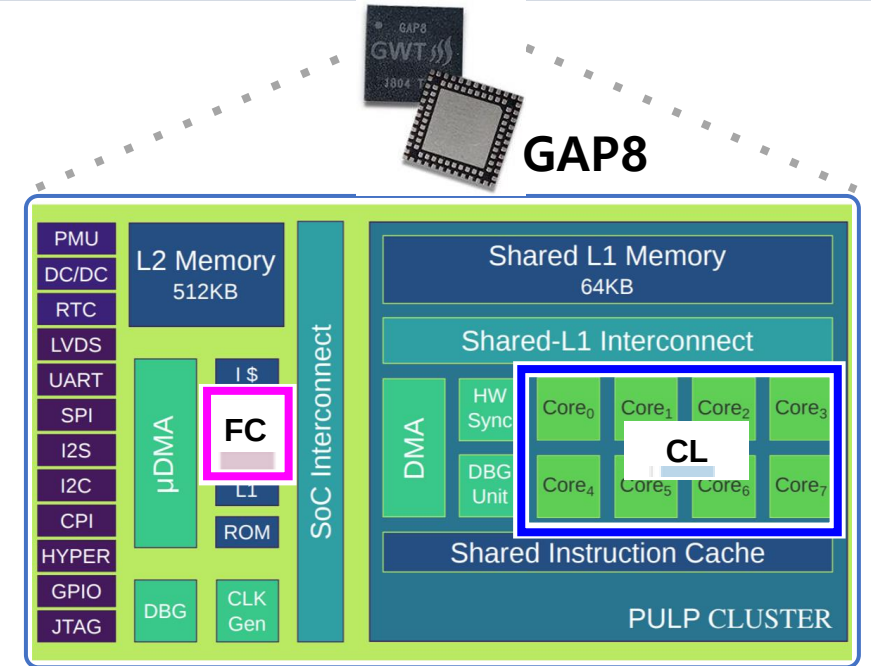
- **Smaller size:** more CNNs can fit in the memory
- **Lower MAC:** higher throughput (faster reactivity)



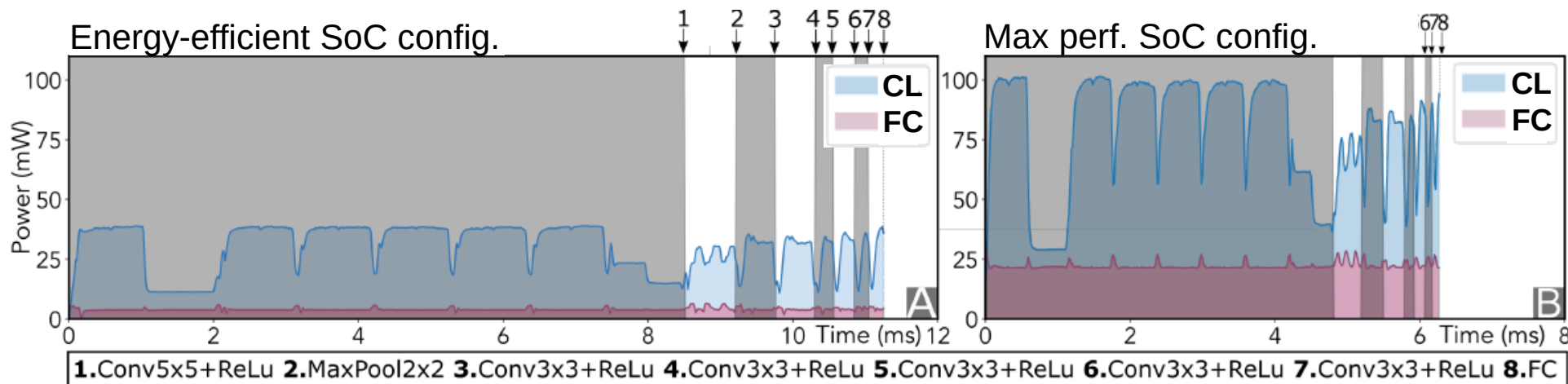
Results: real HW performance

Smallest Tiny-PULP-Dronet inference:

1MMAC, taking 6ms-11ms (depending on SoC config)



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Smallest Tiny-PULP-Dronet inference:

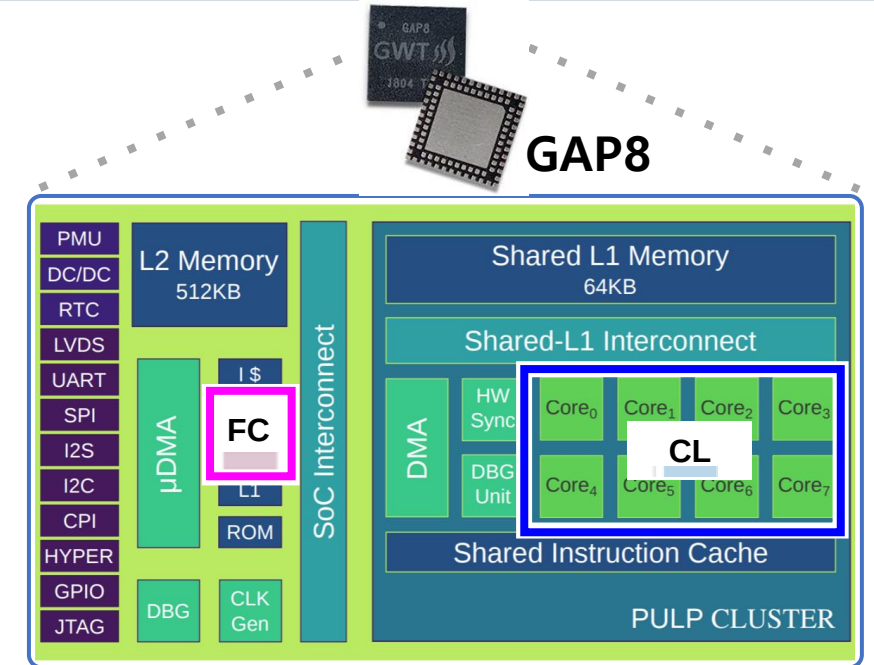
1MMAC, taking 6ms-11ms (depending on SoC config)

Result: improved throughput

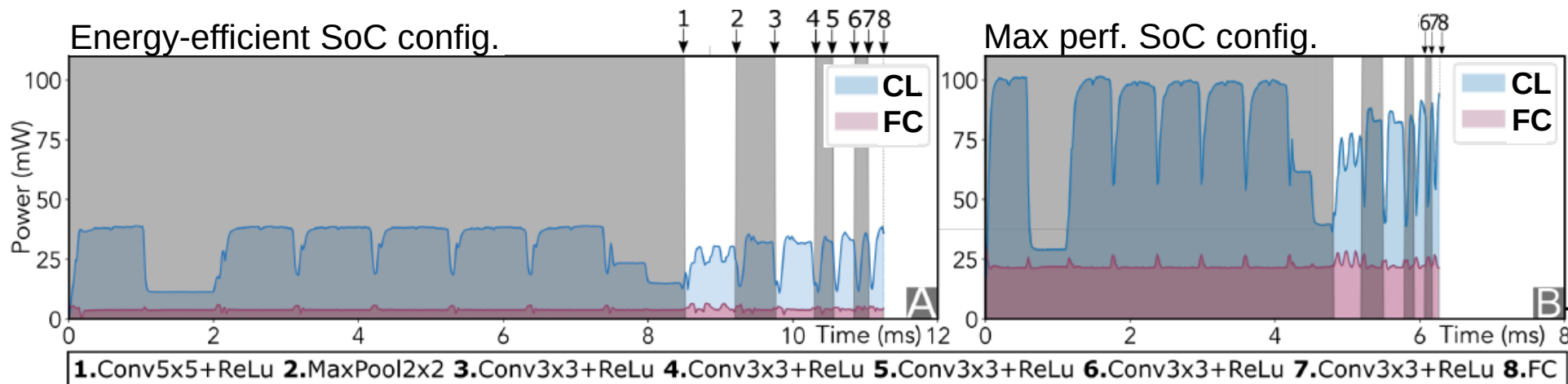
from 10-19 frame/sec to **89-160 frame/s** ↑

→ drone's reactivity **increased by 8.5x.**

(Baseline vs.
Tiny)



D. Palossi: "A 64-mw dnn-based visual navigation engine for autonomous nano-drones"



Real HW power
consumption:
34mW

Conclusion

Nano-UAVs are asking for increased neural intelligence running onboard.

We would like to add many tasks running aboard to tackle more complex flight missions.

We presented **a general methodology** for squeezing CNNs.
We applied it to our PULP-Dronet use case:

- Reduced **memory footprint 50x**
- Reduced **number of operation 27x**
- Increased **throughput by 8.5x**

Outcome: we paved the way toward DNN-based multi-tasking intelligence aboard nano-UAVs.





AICAS 2022



Thank you for your attention !

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